PRODUCT REQUIREMENTS DOCUMENT (PRD) & SYSTEM SPECIFICATION

EDU360 — K–12 Online Academy Outcomes & Active Intelligence Platform

Document Version: 6.0 (Pure Online Schooling & Active Intelligence Edition)
Target Environment: 100% Pure Online Virtual Schooling / Hybrid Virtual K–12 Academies
Canonical Scope: The 8 Canonical Institutional Roles (Zero Physical Transport / Inventory Overhead)
Core Monorepo Stack: Next.js 16.4.2 (App Router / Turbopack) + Python 3.11+ FastAPI (Async ASGI Monolith) + Relational Database

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1. Executive Summary & Product Vision

Traditional School ERPs (such as Fedena, Entab, Teachmint, and Lead School) operate as **passive digital filing cabinets**. They record attendance after the fact, archive marks cards in disconnected tables, issue generic receipts, and inundate parents with unread WhatsApp broadcasts. In a pure online schooling environment, where teachers and learners are separated across digital screens, passive ERPs fail completely because they cannot diagnose why an online student is disengaged, falling behind, or missing homework.

EDU360 is engineered as an **Active Intelligence & Outcomes Engine** specifically optimized for virtual K–12 academies. It unifies Administration, Academic Governance, Examinations, Accounts, Admissions, Faculty, Students, and Parents into a closed-loop real-time ecosystem governed by three core philosophies:

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│                          THE 3 CORE OUTCOME PHILOSOPHIES                     │
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│ 1. Actionable Governance │ 2. Outcome-Oriented Parent│ 3. Zero Extra Clerical │
│    Intelligence          │    Co-Tracking           │    Burden for Faculty  │
│ Dynamic Red/Orange/Green │ Anti-spam moving carousel │ 10-minute auto-attendance,│
│ Triage Engine pinpoints  │ banner + 1-Week Shared    │ in-browser SpeedGrader with│
│ learning drops, attendance│ Focus Sprint table with  │ digital red pen, and   │
│ lags, and SLA bottlenecks│ 10-minute nightly rough-work│ 1-click syllabus pacing │
│ in real time.            │ verification.            │ sync.                  │
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2. Competitive Landscape & Market Differentiation

How EDU360 Dominates Traditional School ERPs & LMS Tools

Functional Vector	Traditional School ERPs (Fedena, Entab, Campus365)	Classroom LMS Tools (Google Classroom, MS Teams)	EDU360 Outcomes Platform (Online Academy)
System Philosophy	Passive relational database; retrospective data entry.	Video link distribution and raw file repository.	**Active Outcomes Engine**: Prescribes micro-interventions and tracks remediation in real time.
Operational Scope	Bloated with physical campus logistics (buses, physical stores).	Pure video & assignment turn-in; zero school administration.	**100% Pure Online Schooling**: Focused strictly on the 8 academic and governance roles.
Academic Remediation	Static marks sheets; remediation left entirely to off-system tutor.	None; only displays grades.	**Automated 1-Week Remedial Focus Sprint**: Auto-enrolls students $\le 70\%$, guiding 7-day micro-steps with physical rough notebook checks.
Homework Evaluation	Download PDF $\to$ open external tool $\to$ re-upload.	Text submission with generic text comments.	**In-Browser SpeedGrader Canvas**: Digital red-pen markup, variable stroke smoothing, and rubric deduction tags.

Parent Engagement	10–15 intrusive WhatsApp/ SMS spam alerts daily.	Email digests that parents routinely ignore.	**Anti-Spam Moving Slider**: Interactive top carousels + high-signal notifications reserved only for verified academic flags.
Live Attendance	Manual roll-call eating 15 minutes of live class time.	Meeting participant CSV export.	**10-Minute Auto-Attendance Rule**: Dynamic join webhook marks Present ≤ 10m, Late > 10m, or Absent automatically.
SLA & Task Governance	Tasks get buried in emails; faculty face penalties without relief.	Basic task assignment without institutional workflow.	**Human-in-the-Loop Governance**: Offers +12h to +72h extensions or peer rebalancing before Principal escalation.
Student School Exit	Disorganized paper clearance forms leading to lost records.	None.	**Dual-Signature Legal Clearance**: Parent digital OTP sign-off + physical certificate scan locked permanently in SIS vault.

3. Platform Strategy for Pure Online Schooling (The 8 Canonical Roles)

EDU360 eliminates traditional physical campus overheads (physical bus transport, uniform inventories, physical laboratory stockrooms) and concentrates all engineering on the **8 Canonical Virtual Academy Roles**:

Stakeholder Role	Primary Modality	Secondary Modality	Role Purpose in Virtual Schooling
1. Super Admin / Principal (`/admin`)	Web App (Desktop / Laptop)	Mobile Web (Executive Alerts)	Full institutional oversight, statutory compliance audit trail, financial health, and high-priority cross-department triage.
2. Academic Coordinator (`/academic_coordinator`)	Web App (Desktop)	Tablet Web	Curriculum pacing surveillance, dynamic timetable substitution cover, faculty SLA workload relief queues.
3. Exam Coordinator (`/exam_coordinator`)	Web App (Desktop)	Tablet Web	Exam schedule planning, blueprint schema locks, accounts fee-clearance gating, and paper moderation distribution.
4. Accounts Coordinator (`/accounts_coordinator`)	Web App (Desktop)	Tablet POS Counter	Real-time fee arrears aging buckets, fee waiver adjustments, bank reconciliation, and exam clearance overrides.
5. Admissions Coordinator (`/admissions_coordinator`)	Web App (Desktop / CRM)	Mobile App (Calling Dialer)	Virtual lead capture, call outcome logging, follow-up scheduling, and atomic 1-click conversion into active SIS student profiles.
6. Teacher (`/teacher`)	Web App (Desktop / Tablet)	Mobile Web	10-Minute auto-attendance monitoring, in-browser SpeedGrader evaluation, 1-Week Remedial Focus Table management, and syllabus delivery logging.
7. Student (`/student` & `/junior`)	Web App (Desktop / Tablet)	Mobile App	Live online class attendance, homework submission, AI Socratic Doubt Tutor, remedial drills, and gamified Junior K–5 adventure quests.
8. Parent (`/parent`)	Mobile-Optimized Web / Mobile App	Web Portal	10-Minute nightly rough-book co-tracking sign-off, live online class attendance radar, fee payments with instant PDF receipts, and anti-spam announcements.

4. Comprehensive 8-Role RBAC & Permissions Matrix

> **SECURITY INVARIANT — SIS Official Document Vault Lock**: Official student legal records (Birth Certificates, Transfer Certificates, Study Conduct Certificates) can **ONLY** be uploaded, sealed, or deleted by **Super Administration**. All other roles have strict **View/Download-Only** permissions.

Module / System Capability	Super Admin	Academic Coord	Exam Coord	Accounts Coord	Admissions Coord	Teacher	Student	Parent
SIS Official Document Vault	Full (Seal/ Edit)	View Only	View Only	View Only	Create Intake	View Only	View Only	View Only
Admissions CRM & SIS Conversion	Full Access	View Only	No Access	Invoicing Sync	Full Access	No Access	No Access	No Access
Executive Priority Triage (Red/Orange/ Green)	Full Access	Academic Scope	Exam Scope	Finance Scope	CRM Scope	Class Scope	No Access	No Access
Curriculum Pacing & Syllabus Delivery	Audit Only	Full Oversight	Blueprint Sync	No Access	No Access	Log Delivery	View Syllabus	View Progress
Timetable & Dynamic Substitution Engine	Audit Only	Full Assign	No Access	No Access	No Access	Accept Cover	View Timetable	View Timetable
Live Online Classes & Attendance Webhooks	Monitor / View	Supervise	No Access	No Access	No Access	Host & Record	Join & Watch	View Timetable
10-Minute Auto-Attendance Engine	Full Override	Department View	Room Audit	No Access	No Access	Auto-Capture/ Edit	View Own Log	View + Leave
SpeedGrader Inline Canvas Markup	View / Audit	Audit Sample	View Papers	No Access	No Access	Full Annotate	View Graded	View Remarks
1-Week Remedial Focus Sprint ($ \le 70\%$)	Institutional View	Department View	Trigger Exam	Clearance Sync	No Access	Prescribe/ Verify	Execute Drills	Nightly Co-Track
Exams, Blueprints & Hall Ticket Locks	Final Lock	Review Sync	Full Authority	Clearance Lock	No Access	Grade Papers	Attend / View	View Results
Fee Milestones, Aging Ledgers & Receipts	Full Authority	No Access	Hall Ticket Gating	Full Authority	Intake Invoice	View Context	View Status	Pay + Receipt
Faculty Workload Relief & Human SLA	Final Escalation	Approve/ Migrate	No Access	No Access	No Access	Request Relief	No Access	No Access
Dual-Signature Legal Exit Clearance	Verify / Archive	Clear Academic	Clear Exam	Clear Dues	No Access	No Access	View Status	Digital Sign-off
In-App Notifications Engine & Deep-Links	Full Broadcast	Department Scope	Exam Scope	Billing Alerts	CRM Alerts	Class Alerts	Learner Alerts	Family Alerts

5. Exhaustive Feature-by-Feature Breakdown (8 Roles)

5.1 Role 1: Super Admin / Principal (`/admin`)

Standard / Normal Features & Workings:

- **Resilient Dashboard Data Hydration**: Loads administrative tasks, triage flags, admissions enquiries, and document queues via `Promise.allSettled`. If any single endpoint times out, remaining widgets render without zero-wiping the UI.
- **CBSE & Regulatory Statutory Audit Logging**: Records every login, grade override, TC creation, and fee adjustment in an append-only audit stream with cryptographic checksums (`GET /api/admin/audit-logs/export/{filename}`).
- **Annual Academic Promotion & Rollover Engine**: 1-Click batch promotion transitioning students from one grade to the next (e.g. Class 8-B $\to$ Class 9-B) while sealing historical transcripts.
- **Dual-Signature Exit Archive**: Stores parent digital OTP signatures alongside high-resolution physical scans of issued Transfer Certificates.

AI / Intelligent Automation Systems:

- **Executive Priority Triage Engine (Red / Orange / Green)**:
- *Mechanism*: Background evaluation running across 5 vectors: Attendance (3+ consecutive absences triggers RED), Academic Drops ($\ge 15\%$ drop triggers RED), Fee Arrears (30+ days overdue triggers ORANGE), Faculty SLA expirations (ORANGE/RED), and Daily Attendance Complete (GREEN).
- *Status*: **100% Real Production Logic** (`backend/app/routes/admin.py`).

In-App Notification Scenarios:

- `ADMIN_EMERGENCY_ESCALATION`: Triggered when an academic task exceeds the +72h grace window without coordinator resolution.
- `ADMIN_UNRESOLVED_ABSENCE_RED`: Triggered when a student crosses 3 consecutive unexcused online class absences.
- `ADMIN_CRITICAL_ARREARS_OVERDUE`: Alerts Super Admin when institution-wide 30+ day unpaid fee arrears cross defined budget thresholds.

5.2 Role 2: Academic Coordinator (`/academic_coordinator`)

Standard / Normal Features & Workings:

- **Curriculum Pacing Surveillance**: Tracks syllabus completion percentages across subjects and sections against scheduled academic calendar weeks.
- **Faculty Master Roster**: Views assigned periods, subject specializations, and daily teaching workloads.
- **Remedial Focus Table Audit**: Monitors Class 8-B and other sections for compliance with the 7-day remedial protocol.

AI / Intelligent Automation Systems:

- **AI Timetable Governor & Dynamic Substitution Allocator**:
- *Mechanism*: When a teacher reports sick or absent, the allocator checks the master timetable matrix, isolates free faculty in that specific period, matches subject compatibility, and badges a `SUBSTITUTION_COVER` period in the substitute's schedule.
- *Status*: **100% Real Algorithmic Constraint Satisfaction Engine** (`backend/app/routes/academic_coordinator.py`).
- **Human-in-the-Loop Faculty Workload Relief Engine**:
- *Mechanism*: Teachers submit relief requests on high-workload days. Academic Coordinators can 1-Click grant extensions (+12h, +24h, +48h, +72h), rebalance tasks to available peer teachers, or escalate with mandatory written justification.
- *Status*: **100% Real State Machine**.

In-App Notification Scenarios:

- `ACADEMIC_SUBSTITUTION_PENDING`: Triggered instantly when a teacher leave request is filed, alerting coordinator to approve substitution slot.
- `ACADEMIC_SYLLABUS_LAG_ALERT`: Triggered when a subject's completed units drop $\ge 15\%$ behind the term pacing schedule.
- `ACADEMIC_RELIEF_REQUEST_FILED`: Alerts coordinator that a faculty member requested SLA clock pause on remedial evaluations.

5.3 Role 3: Exam Coordinator (`/exam_coordinator`)

Standard / Normal Features & Workings:

- **Examination Blueprint Schema Lock**: Defines question structures, marks weightages, and cognitive domains (Recall, Application, Analysis). Blueprints are locked 72 hours before exams.
- **Online Invigilation Roster**: Allocates faculty to virtual proctoring rooms and monitors student webcam feeds during term

exams.
- **Paper Moderation Distribution**: Routes scanned student answer scripts to respective subject teachers with automated grading SLAs.

AI / Intelligent Automation Systems:

- **Accounts-Gated Hall Ticket Lock Engine**:
- *Mechanism*: Exam hall ticket generation queries the Accounts ledger in real time. Defaulters lacking institutional clearance have hall ticket downloads locked until the bursar issues an approval or the invoice is settled.
- *Status*: **100% Real Integrated Policy Gate**.
- **Multimodal Exam Paper Auto-Scanner (SpeedGrader Pre-Check)**:
- *Mechanism*: Utilizes Google Gemini 1.5/2.0 Flash Vision API (`POST /api/exam_coordinator/pre-scan`). Inspects student handwritten scripts against the blueprint key, flagging calculation jumps, sign errors, and formula omissions.
- *Fallback*: If Gemini API is unreachable, provides deterministic heuristic rubric suggestions based on question point allocations.

In-App Notification Scenarios:

- `EXAM_BLUEPRINT_FREEZE_WARNING`: Dispatched 72 hours prior to scheduled examination to lock question paper blueprints.
- `EXAM_GRADING_SLA_NEAR_BREACH`: Dispatched to teacher and exam coordinator when answer script evaluations approach the 48-hour post-exam deadline.
- `EXAM_HALL_TICKET_CLEARANCE_UPDATE`: Real-time notification when a fee waiver unlocks student exam hall ticket access.

5.4 Role 4: Accounts Coordinator (`/accounts_coordinator`)

Standard / Normal Features & Workings:

- **Real-Time Arrears Aging Ledger**: Dynamically categorizes unpaid fee balances into 0–15 Days, 16–30 Days, and 30+ Days buckets.
- **Virtual Counter & Bank Reconciliation**: Logs offline bank transfers (NEFT/RTGS/Cheque) and generates an immutable UTR tracking reference.
- **Management Hardship Waivers & Concessions**: Applies authorized fee discounts with mandatory audit reason codes without deleting historical billing records.
- **Institutional Exam Clearance Override**: Toggles `exam_clearance = True` for financially distressed students without forgiving outstanding liabilities.

AI / Intelligent Automation Systems:

- **Instant Simulated Payment Gateway & Cryptographic Receipt Generator**:
- *Mechanism*: Simulates realistic 2-second bank network latency, processes authorization webhooks, updates ledger balances, and generates a tamper-evident PDF receipt with verification QR code.
- *Status*: **100% Real Production Simulation & PDF Engine**.

In-App Notification Scenarios:

- `ACCOUNTS_NEW_FEE_OVERDUE`: Dispatched when an installment crosses the grace cutoff without receipt.
- `ACCOUNTS_WAIVER_PROCESSED`: Dispatched to student and parent when a fee concession or exam clearance override is granted.
- `ACCOUNTS_ADMISSION_INVOICE_GENERATED`: Triggered when an admissions lead is converted to SIS, prompting counter payment.

5.5 Role 5: Admissions Coordinator (`/admissions_coordinator`)

Standard / Normal Features & Workings:

- **Virtual Admissions Intake CRM**: Captures enquiries, leads, and student intake documents across structured stages (`New Lead` $\to$ `Follow-Up` $\to$ `Virtual Tour` $\to$ `Entrance Cleared` $\to$ `Converted to SIS`).
- **Telephonic Call Outcome Logging**: Records call transcripts, assigns lead temperature (Hot, Warm, Cold), and schedules automated next-touch dates.
- **Statutory Drop Protection**: Inactive leads can only be archived with a mandatory recorded reason (`fees_high`, `curriculum_fit`, `distance`).

AI / Intelligent Automation Systems:

- **Atomic 1-Click SIS Conversion Engine (`POST /api/admin/crm/enquiries/{id}/convert`)**:
- *Mechanism*: Executes an atomic database transaction that creates the permanent `Student` record, assigns roll number and section, promotes intake documents to the official SIS vault, and creates an admission fee invoice in the Accounts ledger.
- *Status*: **100% Real Transactional Workflow**.

In-App Notification Scenarios:

- `ADMISSIONS_LEAD_SLA_BREACH`: Dispatched when a new online lead remains untouched for $> 24\text{ hours}$.
- `ADMISSIONS_CONVERSION_SUCCESS`: Dispatched to Admin, Academic Coordinator, and Accounts when an enquiry is converted to SIS.

5.6 Role 6: Teacher Portal (`/teacher` & `/teacher/focus/[id]`)

Standard / Normal Features & Workings:

- **Subject-Scoped Focus Table**: Mathematics (Mrs. Ananya Verma) filtered by sections (`Class 8-B`, `Class 8-A`, `Class 9-A`, `Class 9-B`, `All`).
- **In-Browser SpeedGrader Canvas**: Teachers evaluate student worksheets and exam scripts directly in the browser with digital red-pen drawing, variable stroke weights, checkmarks, crosses, and rubric deduction markers.
- **1-Click Syllabus Delivery Logging**: Teachers log completed curriculum units with one tap, automatically updating Academic Coordinator pacing charts.

AI / Intelligent Automation Systems:

- **10-Minute Online Class Auto-Attendance Rule**:
- *Mechanism*: Evaluates live class join timestamps via FastAPI webhook. ≤ 10 minutes = Present; $10-25$ minutes = Late (1-click approval); > 25 minutes = Absent.
- *Status*: **100% Real Attendance Logic**.
- **The 1-Week Remedial Focus Sprint Engine**:
- *Mechanism*: Students scoring $\le 70\%$ in Mathematics exams are auto-enrolled in 7-day remedial sprints. Gemini 1.5/2.0 API (`POST /api/remedial/generate-sprint`) analyzes exam errors and generates daily micro-targets.
- *State Machine Invariant*: Past days ($d < \text{current day}$) can never be "Upcoming"—only `Completed` ✓ or `Pending Review` ⏳.
- *Fallback*: If Gemini API is offline, an algorithmic pedagogical template generates the 7-day micro-syllabus.
- **Rough Work Notebook AI Diagnostic**:
- *Mechanism*: Analyzes photo uploads of student physical rough notebook working alongside the assigned question and rubric checkpoint.
- *Status*: **Gemini Vision Prompt with Heuristic Fallback Checkpoint Evaluator**.

In-App Notification Scenarios:

- `TEACHER_REMEDIAL_SUBMISSION_PENDING`: Dispatched when a student uploads Day N rough work, prompting teacher review.
- `TEACHER_SUBSTITUTION_ASSIGNED`: Alerts teacher that an emergency cover period has been injected into their timetable.
- `TEACHER_SPEEDGRADER_DEADLINE`: Dispatched 6 hours before homework evaluation cutoff.

5.7 Role 7: Student Portals (Secondary `/student` & Junior K–5 `/junior`)

Standard / Normal Features & Workings:

- **Secondary Student LMS**: Daily timetable, 1-click video class join, categorized recorded class archive (Subject/Chapter/Date), and homework upload dropzone.
- **Junior K–5 Adventure Quest Portal**: Touch-first gamified interface featuring an adventure map timetable, Magic Booster chests containing formula cards, and daily academic quests.
- **Virtual Class Presence Radar**: Replaces physical bus radar with real-time class connectivity, audio/video test status, and upcoming live session countdowns.

AI / Intelligent Automation Systems:

- **AI Socratic Doubt Tutor (`POST /api/student/ai-tutor`)**:
- *Mechanism*: Context-aware AI tutor powered by Gemini. Guides secondary students step-by-step through mathematical concepts without giving direct answers, encouraging critical thinking.
- *Fallback*: Deterministic Socratic hint bank categorized by Class 8 Mathematics chapters.

- **15-Minute Missed-Cutoff Homework Backlog Engine**:
- *Mechanism*: Background asyncio scheduler checks unsubmitted homework past strict cutoff time (e.g. 5:00 PM). Transitions status from `PENDING` to `BACKLOG` and generates a unique forensic evidence ID `#EVD-BKLOG-XXX`.
- *Status*: **100% Real Background Engine**.

In-App Notification Scenarios:

- `STUDENT_HOMEWORK_GRADED`: Alerts student when teacher completes SpeedGrader annotations on their worksheet.
- `STUDENT_REMEDIAL_DAY_UNLOCKED`: Dispatched when Day N rough work is approved, unlocking Day $N+1$.
- `STUDENT_LIVE_CLASS_10MIN_REMINDER`: Dispatched 10 minutes prior to live online class start.

5.8 Role 8: Parent Portal (`/parent`)

Standard / Normal Features & Workings:

- **Anti-Spam Moving Slider Carousel**: Routine school announcements and circulars glide across a top banner carousel, keeping the notification tray clean and high-signal.
- **10-Minute Nightly Rough-Book Co-Tracking**: Parents review their child's daily remedial math target, physically inspect their rough notebook, and sign off with a single tap.
- **Simulated Instant Fee Checkout**: 2-second bank authorization that settles fee invoices and produces a digitally sealed PDF receipt.

AI / Intelligent Automation Systems:

- **AI Parent Study Companion**:
- *Mechanism*: Explains tonight's remedial concept in simple, encouraging language so parents without advanced math backgrounds can support their child effectively.
- *Fallback*: Pre-compiled parental coaching prompts for Class 8 Mathematics.

In-App Notification Scenarios:

- `PARENT_REMEDIAL_NIGHTLY_SIGN_OFF`: Dispatched at 7:00 PM reminding parent to verify child's 10-minute rough work.
- `PARENT_CONSECUTIVE_ABSENCE_ALERT`: High-priority alert triggered when child misses 2+ consecutive live classes.
- `PARENT_FEE_INSTALLMENT_DUE`: Dispatched 3 days prior to fee milestone due date.

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6. The 13 Active AI & Intelligent Systems (Comprehensive Technical Index)

The table below provides a complete technical audit of every automated and AI-powered feature in the EDU360 codebase, distinguishing real working production logic from heuristic fallback modes:

#	Intelligent System Name	Primary Role	Operational Mechanism	AI Prompt / Logic Specification	Real Working vs Fallback Mode
1	**1-Week Remedial Focus Sprint Engine**	Teacher / Student	Generates 7-day micro-target curriculum for students scoring $ \le 70\%$ in Mathematics.	Multimodal prompt taking student weak concepts, past errors, and textbook chapter syllabus.	**Real**: Gemini API call (`remedial.py`). **Fallback**: Algorithmic pedagogical template.
2	**Rough Work Pre-Verification Diagnostic**	Teacher / Parent	Inspects photo upload of student physical rough notebook calculations against teacher rubric.	Multimodal Vision prompt detecting calculation skips, sign distribution errors, and missing steps.	**Real**: Gemini Vision API. **Fallback**: Rule-based checkpoint compliance scorer.
3	**Multimodal Exam Paper Auto-Scanner**	Exam Coord / Teacher	Pre-scans uploaded answer scripts, drafts score suggestions, and highlights suspected errors.	Gemini Vision prompt mapping handwritten steps to question blueprint marks distribution.	**Real**: Gemini 1.5/2.0 API. **Fallback**: Mock grading schema with manual canvas edit.

4	**AI Socratic Doubt Solver / Tutor**	Student (Secondary)	Academic chat assistant answering student doubts through iterative hints without direct answers.	Socratic instructional system prompt: "Never reveal final answer; ask guided leading questions."	**Real**: Gemini Chat API. **Fallback**: Chapter-specific Socratic hint dictionary.
5	**AI Parent Study Companion**	Parent	Translates complex academic math topics into 5-minute actionable coaching tips for parents.	Generative prompt simplifying pedagogical steps into plain, supportive conversational guidance.	**Real**: Gemini API. **Fallback**: Curated parental study tips library.
6	**Executive Priority Triage Engine**	Super Admin	Runs every 60s evaluating Attendance, Academic Drops, Fee Arrears, and SLA Tasks across Red/ Orange/Green.	Deterministic multi-vector rule evaluator: 3+ absences = RED; $\ge 15\%$ section drop = RED; 30d arrears = ORANGE.	**100% Real Production Code** (`backend/app/routes/admin.py`).
7	**10-Minute Auto-Attendance Engine**	Teacher / Admin	Webhook listener comparing student class join timestamp against scheduled live session start time.	$\Delta T = T_{\text{join}} - T_{\text{start}}$. ≤ 10m: Present, $10-25$m: Late (1-click approve), >25m: Absent.	**100% Real Production Code** (`live_classes.py`).
8	**15-Minute Homework Backlog Engine**	Academic Coord / Teacher	Scheduled background worker monitoring unsubmitted assignments past the daily submission cutoff.	Sweeps active homework records every 15 minutes; transitions expired to `BACKLOG` + generates `#EVD-BKLOG-XXX`.	**100% Real Background Engine** (Asyncio background loop).
9	**AI Timetable Governor & Substitutions**	Academic Coord / Teacher	Matches absent teacher periods with available, non-busy faculty qualified in compatible subjects.	Algorithmic constraint solver checking daily timetable matrix, free slots, and subject tags.	**100% Real Algorithmic Engine** (`academic_coordinator.py`).
10	**Faculty Workload Relief State Machine**	Teacher / Academic Coord	Allows teachers to pause SLA timers on heavy days; enables coordinators to grant extensions or migrate tasks.	State machine: `ACTIVE` $\to$ `RELIEF_REQUESTED` $\to$ `EXTENSION_GRANTED` or `MIGRATED` with 0 faculty penalty.	**100% Real State Machine** (`tasks.py`).
11	**Curriculum Pacing & Lag Diagnostic**	Academic Coord / Teacher	Calculates percentage delivery of curriculum units against academic calendar week milestones.	Pacing formula: $\text{Lag} = \text{Scheduled Units} - \text{Delivered Units}$; flags $> 15\%$ delay as department warning.	**100% Real Mathematical Logic**.
12	**Atomic Admissions CRM SIS Converter**	Admissions / Accounts	1-Click atomic transaction promoting lead to SIS student, creating roll number, and creating fee ledger entry.	Transactional pipeline: Lead $\to$ Student record $\to$ Document vault promote $\to$ Accounts invoice generation.	**100% Real Database Transaction** (`admin.py`).
13	**Dual-Signature Legal Exit Clearance**	Admin / Parent	Multi-party clearance workflow combining parent digital OTP sign-off with scanned physical document upload.	Cryptographic verification tying user session OTP with high-res document image hash locked in SIS vault.	**100% Real Multi-Party Workflow**.

7. In-App Notifications Engine & Event-Bus Architecture

The In-App Notification Engine operates as a central event bus serving all 8 canonical online roles.

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│                            CENTRAL EVENT BUS PIPELINE                        │
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│ 1. Event Emitter (FastAPI Routes / Background Workers)                       │
│    -> Emits typed event: (user_id, role, title, message, type, action_url, priority)   │
│                                                                              │
│ 2. Deduplication & Throttling Filter                                         │
│    -> 5-Minute Deduplication Window prevents spamming duplicate alerts to same user    │
│                                                                              │
│ 3. Database Persistence (Table: notifications)                               │
│    -> Indexed with composite B-Tree: ix_notifications_user_unread_sort (< 2ms query)   │
│                                                                              │
│ 4. Client Consumption (Next.js Header Badge & Dropdown)                      │
│    -> SWR / React Query polling (30s) + optimistic read toggle + deep-link navigation  │
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```

Notification Database DDL

```
CREATE TABLE notifications (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER REFERENCES users(id) ON DELETE CASCADE,
    role VARCHAR(50) NOT NULL,
    title VARCHAR(150) NOT NULL,
    message TEXT NOT NULL,
    type VARCHAR(50) NOT NULL, -- e.g. remedial_due, attendance_red, fee_overdue
    action_url VARCHAR(255),  -- contextual deep link (e.g. /teacher/focus/8B?student=12)
    is_read BOOLEAN DEFAULT 0,
    priority VARCHAR(20) DEFAULT 'medium', -- low, medium, high, urgent
    channel VARCHAR(30) DEFAULT 'in_app',
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);
-- High-Performance Composite Indexes
CREATE INDEX ix_notifications_user_unread_sort ON notifications(user_id, is_read, created_at DESC);
CREATE INDEX ix_notifications_role_unread_sort ON notifications(role, is_read, created_at DESC);
CREATE INDEX ix_notifications_cleanup ON notifications(created_at);
```

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8. Key System Workflows & Flow Diagrams

Workflow 8.1: Admissions Intake → Follow-up → Atomic SIS & Fee Conversion

Workflow Logic Flowchart

```
flowchart TD
    A["Virtual Parent Enquiry Received"] --> B["Admissions Lead Created (Stage: NEW)"]
    B --> C["Admissions Coordinator Virtual Outreach Call"]
    C --> D{"Parent Interest Level?"}
    D -->|Interested / Tour Booked| E["Schedule Virtual Academy Tour & Document Intake"]
    D -->|Undecided| F["Set Follow-up Date (Stage: FOLLOWUP)"]
    D -->|Unresponsive / Drop| G["Mark DROPPED + Log Mandatory Statutory Reason"]
    E --> H["Virtual Entrance Evaluation Cleared"]
    H --> I["1-Click Convert to SIS Triggered"]
    I --> J["Generate Unique Roll & Admission ID (EDU-2026-XXX)"]
    I --> K["Promote Documents to Official SIS Vault"]
    I --> L{"Admission Fee Settled?"}
    L -->|Yes| M["Create Reconciled Payment Record (UTR Generated)"]
    L -->|No / Partial| N["Create FeeDefaulter Record (is_admission_fee = True)"]
    J --> O["Student Active in All 8 Online Portals"]
    M --> O
    N --> O
```

Workflow 8.2: 10-Minute Online Class Auto-Attendance & Absence Escalation

Workflow Logic Flowchart

```
flowchart TD
    A["Teacher Launches Live Video Class (T = 0 mins)"] --> B["FastAPI Webhook Listener Activated"]
```

```
    B --> C["Student Joins Live Video Session"]
    C --> D{"Join Timestamp (T)"}
    D -->|T <= 10 mins| E["Auto-Marked: PRESENT in SIS"]
    D -->|10 mins < T <= 25 mins| F["Flagged: LATE (Teacher 1-Click Acknowledge)"]
    D -->|T > 25 mins or No Join| G["Auto-Marked: ABSENT in SIS"]
    G --> H{"Absence History Evaluation"}
    H -->|Consecutive Absences < 3 Days| I["Attendance Updated in Parent Daily Portal"]
    H -->|Consecutive Absences >= 3 Days| J["CRITICAL RED ALERT on Admin Dashboard"]
    J --> K["Automated High-Priority WhatsApp/In-App Notice to Parent"]
    J --> L["Mandatory Student Care Case Assigned to Class Teacher"]
```

Workflow 8.3: Examination Paper Upload → AI Pre-Scan → SpeedGrader Moderation

Workflow Logic Flowchart

```
flowchart TD
    A["Term Exam Conducted & Papers Uploaded"] --> B["Exam Submissions Repository"]
    B --> C["FastAPI Multimodal AI Diagnostic Engine (Gemini)"]
    C --> D["AI Scans Step-by-Step Working against Teacher Rubric"]
    D --> E["Flag Suspected Errors (e.g. Sign Distribution in Step 2)"]
    E --> F["Teacher Opens In-Browser SpeedGrader Canvas"]
    F --> G["Canvas Pre-loaded with Student Script + Tentative AI Flags"]
    G --> H["Teacher Applies Digital Red-Pen Markup & Finalizes Score"]
    H --> I["Grade Locked & Synchronized to SIS Marks Record"]
    I --> J{"Final Exam Score <= 70%?"}
    J -->|Yes (<= 70%)| K["Auto-Enrolled in 1-Week Remedial Focus Sprint"]
    J -->|No (> 70%)| L["Mastery Baseline Satisfied (Exempt from Focus Sprint)"]
```

Workflow 8.4: The 1-Week Remedial Focus Sprint & Rough Work Co-Tracking

Workflow Logic Flowchart

```
flowchart TD
    A["Student Enrolled in Focus Table (Subject: Mathematics)"] --> B["Gemini Generates 7-Day Targeted Practice Sprint"
    B --> C["Day 1: Micro-Problem Assigned + Formula Booster Card"]
    C --> D["Student Solves in Physical Rough Notebook & Uploads Photo"]
    D --> E["Teacher Inspects Solution Steps & Rubric Checkpoint"]
    E --> F{"Teacher Review Decision"}
    F -->|Satisfactory| G["Approve & Mark Day Completed ✓"]
    F -->|Needs Revision| H["Return for Revision ⏳ + Actionable Note"]
    G --> I["Nightly Parent 10-Minute Co-Tracking Verification"]
    I --> J["Parent Inspects Physical Rough Book & Taps Sign-Off"]
    J --> K{"Sprint Day Number?"}
    K -->|Day < 7| L["Advance to Day d+1 (Next Micro-Target)"]
    K -->|Day = 7 Completed| M["Remedial Sprint Mastery Certified ✓"]
```

Workflow 8.5: Dynamic Timetable Governor & Substitution Cover

Workflow Logic Flowchart

```
flowchart TD
    A["Teacher Submits Sick Leave Request"] --> B["FastAPI Timetable Governor Triggered"]
    B --> C["Isolate Absent Teacher's Periods for the Day"]
    C --> D["Scan Master Timetable for Available Peer Faculty"]
    D --> E["Filter by Subject Compatibility & Free Slot Matrix"]
    E --> F["Rank Optimal Substitute Faculty"]
    F --> G["Academic Coordinator 1-Click Approves Substitution"]
    G --> H["Badge SUBSTITUTION_COVER in Substitute's Portal"]
    G --> I["Update Student Daily Class Schedule with Cover Teacher"]
```

Workflow 8.6: Human-in-the-Loop Faculty SLA Relief & Rebalancing

Workflow Logic Flowchart

```
flowchart TD
```

```
    A["Teacher Assigned Grading / Remedial Task (48h SLA)"] --> B["Dynamic Countdown Active"]
    B --> C{"Teacher Experiencing Workload Strain or Emergency?"}
    C -->|Yes| D["Teacher Clicks [Request Task Relief / Extension]"]
    D --> E["SLA Clock Paused Immediately (Zero Penalty)"]
    E --> F["Academic Coordinator Review Cockpit Alerted"]
    C -->|No| G{"SLA Timer Nears Expiration?"}
    G -->|Timer Active| B
    G -->|Timer Expired| F
    F --> H{"Academic Coordinator Human Decision"}
    H -->|Option 1| I["[Grant SLA Extension] (+12h, +24h, +48h, +72h)"]
    H -->|Option 2| J["[Migrate Task to Peer Faculty] (Zero Penalty)"]
    H -->|Option 3| K["[Escalate to Principal] (Mandatory Justification)"]
    I --> B
    J --> L["Peer Teacher Notified & Daily Schedule Adjusted"]
    K --> M["Super Admin / Principal Executive Action"]
```

Workflow 8.7: Dual-Signature Student Exit & Transfer Certificate Clearance

Workflow Logic Flowchart

```
flowchart TD
    A["Parent Submits Student Exit / Transfer Request"] --> B["Automated Clearance Audit Initiated"]
    B --> C["Verify Accounts Clearance (Zero Outstanding Arrears)"]
    B --> D["Verify Academic & Exam Clearance (Marks Recorded)"]
    C & D --> E{"All Online School Clearances Granted?"}
    E -->|No| F["Hold Clearance & Alert Respective Department Heads"]
    E -->|Yes| G["Admin Generates Official Transfer Certificate & Bonafide"]
    G --> H["Step 1: Parent Reviews Digital Copy & Submits OTP Sign-off"]
    H --> I["Digital Acceptance Timestamped & Sealed in SIS"]
    I --> J["Step 2: Hardcopy Certificate Handed Over / Couriered"]
    J --> K["Admin Uploads Scanned Acknowledgment Slip to Vault"]
    K --> L["PERMANENT DUAL-SIGNATURE LEGAL LOCK (Zero Dispute Risk)"]
```

---

9. Exhaustive Relational Database Schema & Data DDL

The database engine runs on **Async SQLAlchemy 2.0** with strict **SQLite PRAGMA foreign keys** enabled on every session:

Workflow Logic Flowchart

```
erDiagram
    USERS ||--o{ STUDENTS : "mentors/guards"
    STUDENTS ||--o{ ATTENDANCE_RECORDS : "tracks"
    STUDENTS ||--o{ FEE_MILESTONES : "billed"
    STUDENTS ||--o{ FEE_DEFAULTERS : "owes"
    STUDENTS ||--o{ REMEDIAL_SPRINTS : "enrolled"
    STUDENTS ||--o{ EXAM_SUBMISSIONS : "submits"
    EXAMS ||--o{ EXAM_SUBMISSIONS : "contains"
    EXAMS ||--o{ REMEDIAL_SPRINTS : "triggers"
    USERS ||--o{ ADMIN_TASKS : "assigned"
    STUDENTS ||--o{ OFFICIAL_DOCUMENTS : "stores"
    LIVE_CLASSES ||--o{ ATTENDANCE_RECORDS : "generates"
    USERS ||--o{ NOTIFICATIONS : "receives"
```

Production SQL Schema Definitions

1. Table: `remedial_sprints`

```
CREATE TABLE remedial_sprints (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    exam_id INTEGER REFERENCES exams(id) ON DELETE SET NULL,
    subject VARCHAR(50) NOT NULL,
    exam_score_pct FLOAT NOT NULL,
    current_day INTEGER DEFAULT 1,
    total_days INTEGER DEFAULT 7,
    status VARCHAR(20) DEFAULT 'active', -- active, completed, queued
    weak_concept VARCHAR(150),
```

```
    teacher_verification_notes TEXT,
    parent_acknowledgment BOOLEAN DEFAULT 0,
    priority_rank INTEGER DEFAULT 1,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX ix_remedial_student_status ON remedial_sprints(student_id, status);
CREATE INDEX ix_remedial_subject_score ON remedial_sprints(subject, exam_score_pct);
```

2. Table: `fee_defaulters`

```
CREATE TABLE fee_defaulters (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER REFERENCES students(id) ON DELETE RESTRICT,
    student_name VARCHAR(100) NOT NULL,
    class_name VARCHAR(30) NOT NULL,
    roll_no VARCHAR(20),
    parent_name VARCHAR(100) NOT NULL,
    parent_phone VARCHAR(20) NOT NULL,
    milestone_title VARCHAR(150) NOT NULL,
    amount_due FLOAT NOT NULL,
    days_overdue INTEGER DEFAULT 0,
    reminder_sent BOOLEAN DEFAULT 0,
    reminder_count INTEGER DEFAULT 0,
    last_reminder_at DATETIME,
    is_admission_fee BOOLEAN DEFAULT 0,
    exam_clearance BOOLEAN DEFAULT 0,
    waiver_amount FLOAT DEFAULT 0.0,
    waiver_reason VARCHAR(255),
    waiver_approved_by VARCHAR(100),
    waiver_approved_at DATETIME,
    payment_history_json TEXT,
    notes TEXT,
    status VARCHAR(30) DEFAULT 'Overdue',
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX ix_fee_defaulter_admission_status ON fee_defaulters(is_admission_fee, status);
CREATE INDEX ix_fee_defaulter_aging_status ON fee_defaulters(days_overdue, status);
CREATE INDEX ix_fee_defaulters_student_id ON fee_defaulters(student_id);
```

3. Table: `attendance_records`

```
CREATE TABLE attendance_records (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    class_id INTEGER REFERENCES live_classes(id) ON DELETE SET NULL,
    student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    student_name VARCHAR(100) NOT NULL,
    join_timestamp VARCHAR(50),
    minutes_late FLOAT DEFAULT 0.0,
    status VARCHAR(20) DEFAULT 'Present',
    teacher_override BOOLEAN DEFAULT 0,
    date VARCHAR(30) NOT NULL
);
CREATE INDEX ix_attendance_student_date ON attendance_records(student_id, date);
```

4. Table: `exam_submissions`

```
CREATE TABLE exam_submissions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    exam_id INTEGER NOT NULL REFERENCES exams(id) ON DELETE CASCADE,
    student_id INTEGER NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    status VARCHAR(30) DEFAULT 'pending',
    paper_file_url TEXT,
    paper_file_name VARCHAR(255),
    submitted_at DATETIME,
    marks_obtained FLOAT,
    percentage FLOAT,
    teacher_feedback TEXT,
    graded_at DATETIME,
    graded_by_teacher_id INTEGER REFERENCES users(id) ON DELETE SET NULL,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_exam_student_submission UNIQUE (exam_id, student_id)
);
```

5. Table: `admin_tasks`

```
CREATE TABLE admin_tasks (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    title VARCHAR(150) NOT NULL,
    description TEXT,
    category VARCHAR(50) NOT NULL, -- e.g. academic, compliance, finance, verification
    assigned_role VARCHAR(50) NOT NULL,
    assigned_user_id INTEGER REFERENCES users(id) ON DELETE SET NULL,
    status VARCHAR(30) DEFAULT 'pending', -- pending, in_progress, relief_requested, completed
    priority VARCHAR(20) DEFAULT 'medium',
    due_date DATETIME NOT NULL,
    sla_hours INTEGER DEFAULT 48,
    relief_reason TEXT,
    extension_hours INTEGER DEFAULT 0,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX ix_admin_tasks_role_status ON admin_tasks(assigned_role, status);
```

10. System Architecture, Resilient Login & 203 Endpoints

Monorepo Topology

```
EDU360 System Topology (Pure Online Schooling Monorepo)
├── Frontend Layer: Next.js 16.4.2 (App Router) + React 19 + TypeScript + Tailwind CSS
│   ├── Client State Strategy: SWR / React Query + LocalStorage Resilient Profile Cache
│   ├── SpeedGrader Canvas: HTML5 Canvas Annotation Engine with Digital Red-Pen smoothing
│   ├── Resilient Auth: Universal Password123 + 3s AbortController fallback profile resolver
│   └── Dashboard Hydration: Promise.allSettled preventing UI data-zeroing on network hiccups
│
├── Backend Layer: Python 3.11+ FastAPI (Async ASGI Monolith)
│   ├── Concurrency Model: Uvicorn Async Worker Loop (Serving 203 Endpoints)
│   ├── Background Scheduler: Asyncio Task sweeping 15-minute homework cutoffs & SLA checks
│   └── Multi-Domain Routers (Pure Online Schooling):
│       ├── /api/auth               -> Identity, Resilient Session & Role RBAC
│       ├── /api/admin              -> CRM, Admissions, Dual-Signature Exit, Rollover, Triage
│       ├── /api/academic_coordinator -> Pacing Surveillance, Dynamic Substitutions, Relief
│       ├── /api/exam_coordinator    -> Blueprint Moderation, Hall Tickets, Invigilation
│       ├── /api/finance            -> Defaulter Aging, Reconciled Gateway, Fee Waivers
│       ├── /api/teacher            -> 10-Min Attendance, SpeedGrader Canvas, Focus Table
│       ├── /api/student            -> Homework Uploads, AI Socratic Tutor, Class Recordings
│       ├── /api/parent             -> 10-Min Co-Tracking, Class Radar, Fee Checkout
│       ├── /api/remedial           -> 1-Week Focus Sprint Engine, Rough Work Diagnostic
│       └── /api/notifications      -> Real-Time In-App Alert Bus, Deep-Link Dispatch
│
└── Database & AI Tier:
    ├── Primary Database: SQLite 3 with Async aiosqlite + PRAGMA foreign_keys = ON
    └── AI Diagnostic Engine: Google Gemini 1.5/2.0 Flash Multimodal Vision API
```

11. Phased Deployment & Cloud Scaling Strategy

Workflow Logic Flowchart

```
gantt
    title EDU360 Online Academy Deployment Roadmap
    dateFormat  YYYY-MM-DD
    section Phase 1: Core Cloud Infrastructure
    Database Migration (SQLite -> AWS RDS PostgreSQL 16) :2026-11-01, 7d
    Cloudflare R2 Encrypted Document Blob Storage       :2026-11-05, 5d
    Domain Setup & Wildcard SSL Certification           :2026-11-08, 3d
    section Phase 2: Production Services
    Razorpay / Cashfree Merchant KYC & Webhooks         :2026-11-12, 6d
    Meta WhatsApp Cloud Business API Webhooks           :2026-11-15, 8d
    Firebase Cloud Messaging (FCM) Web Push Dispatch    :2026-11-20, 5d
    section Phase 3: Institutional Pilot
    Faculty SpeedGrader & Focus Sprint Training         :2026-11-25, 7d
    Parent Mobile Web App Rollout (Class 8 Pilot)       :2026-12-02, 10d
    Full Virtual Academy Outcomes Go-Live               :2026-12-15, 5d
```